

A systematic review and meta-analysis suggests obesity predicts onset of chronic kidney disease in the general population



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Obesity and chronic kidney disease (CKD) are public health priorities that share core pathophysiological mechanisms. However, whether high body mass index (BMI) increases risk of CKD *de novo* remains ill-defined. To evaluate the role of BMI in predicting CKD onset in the general adult population, we performed a systematic review and meta-analysis of PubMed and ISI Web of Science databases articles published between January 2000 and August 2016 without language restriction. We selected studies in adult individuals from a general population with normal renal function at baseline that reported the risk of low estimated glomerular filtration (eGFR) (under 60 mL/min/1.73m²) and/or albuminuria (1+ at dipstick or an albumin creatinine ratio of 3.4 mg/mmol or more) as hazard ratio, odds ratio or relative risk related to obesity, overweight, or BMI as continuous value. A total of 39 cohorts covering 630, 677 participants with a mean follow-up of 6.8 years were selected. Obesity increased the relative risk, 95% confidence interval and heterogeneity (I²) of developing low eGFR (1.28, 1.07–1.54, [I²: 95.0%]) and albuminuria (1.51, 1.36–1.67, [I²: 62.7%]). Increase of BMI unit was also associated with higher risk of low eGFR (1.02, 1.01–1.03, [I²: 24.3%]) and albuminuria (1.02, 1.00–1.04, [I²: 0%]). Conversely, overweight did not predict onset of either low eGFR (1.06, 0.94–1.21, [I²: 50.0%]) or albuminuria (1.24, 0.98–1.58, [I²: 49.4%]). Thus, a high BMI predicts onset of albuminuria without kidney failure (CKD stages 1–2) as well as CKD stages 3 and higher, the effect being significant only in obese individuals. Hence, our findings may have implications to improve risk stratification and recommendations on body weight control in the general population.

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Obesity is now considered a global public health challenge, with a growing worldwide prevalence.¹ According to the most recent estimates, age-standardized prevalence of obesity will increase from 11% to 18% in men and from 15% to 21% in women by 2025.²

Besides the worrisome prevalence, obesity markedly increases the risk of developing hypertension, diabetes, and cardiovascular disease.³

Conversely, the association between obesity and chronic kidney disease (CKD) remains controversial. CKD is now recognized as a health priority worldwide due to the negative effects on prognosis and the quality of life of patients, and on the economic resources of countries.^{4–6}

Previous analyses detected the association between higher body mass index (BMI) and risk for end-stage renal disease (ESRD) in the general population. However, in most of these studies (with the exception of the study by Hsu *et al.*⁷), the results were flawed by the inclusion of patients with CKD at baseline and lack of adjustment for a baseline estimated glomerular filtration rate (eGFR)^{7–11}; these 2 aspects represented a relevant bias because presence of CKD *per se* increases the risk of ESRD. Moreover, the several studies that investigated the role of obesity at the onset of overt CKD provided controversial results, with more recent and larger studies suggesting a predictive role^{12–18} that was not confirmed by others.^{19–24} No systematic analysis has formally addressed this issue. Only 1 meta-analysis of 19 cohort studies that involved approximately 6.3 million subjects showed a significant association between obesity and kidney disease²⁵; however, these authors evaluated a highly heterogeneous renal endpoint, that is, a combination of ESRD, kidney stones, positive proteinuria at dipstick, renal cell carcinoma, and incident CKD (eGFR <60 mL/min/1.73 m²). Because this latter endpoint was assessed in only 15,000 subjects enrolled in 2 cohort studies, no definitive conclusion could be drawn on the obesity-related risk of CKD. Therefore, whether high body mass index acts as independent predictor of new-onset CKD currently remains undefined.

To fill this relevant gap of knowledge, we performed a systematic review and meta-analysis of longitudinal cohorts derived from an adult general population, to quantify the predictive role of obesity, overweight, and BMI as continuous variables on the risk of new-onset CKD. This issue is critical to improve risk stratification, and consequently, to optimize

preventive strategies aimed at reducing CKD prevalence in the general population.

RESULTS

Characteristics of studies

After screening titles and abstracts, 106 of 5921 studies were considered (Figure 1). The full text of each article was reviewed by 2 authors, and 39 studies were included in the present meta-analysis. Agreement of 2 reviewers was good for study selection and quality assessment ($\kappa = 0.934$ and 0.849 , respectively). The reasons for exclusion of 67 studies are shown in Figure 1.

Selected studies were grouped in 9 categories based on 3 predictors (obesity, overweight, BMI) and 3 outcomes (isolated low eGFR, isolated albuminuria, and CKD defined by either eGFR <60 ml/min/1.73 m² or albuminuria, whichever was used in a given report). When considering obesity as a predictor, 13 studies reported data on the risk of low eGFR,^{12–24} 7 studies reported on isolated albuminuria,^{13,14,17,19,23,26,27} and 3 studies reported on CKD.^{19,28,29} When considering overweight as a predictor, 8 studies reported data on the risk of low eGFR,^{15,18,20,23,24,30–32} and 2 studies reported on the risk of albuminuria.^{23,26} BMI, as continuous variable, was analyzed in 13 studies that reported data on the risk of low eGFR,^{15,24,33–43} 2 studies reported on the risk of albuminuria,^{44,45} and 3 reported on CKD risk^{46–47} (Table 1). Data on the measured incident rate of low eGFR related to obesity were available in only 1 study.¹⁶ We also found 6 studies that evaluated the risk of low eGFR or CKD among obese subjects with and without metabolic syndrome.^{12,13,48–51}

Selected studies are listed in Table 1. Overall, studies included information on 630,677 individuals. The sample size of these studies ranged from 454 to 123,764 participants. Age range of the selected cohorts was 35 to 76 years. Mean follow-up duration (6.8 years; range: 2.0–18.5), was >5 years in 23 studies and >10 years in 4 studies (Table 1). eGFR was estimated by the Modification of Diet in Renal Disease (MDRD) equation in 32 studies, whereas the remaining used the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula. Twenty-nine studies included Asian Pacific populations ($n = 512,230$ participants), that is, Japan ($n = 16$), Korea ($n = 10$), China ($n = 2$), and Thailand ($n = 1$), whereas 10 studies included Western populations ($n = 118,447$ participants), that is, United States ($n = 7$), Austria ($n = 1$), and Iran ($n = 2$). As described in Table 1, 32 studies adjusted for the main demographic characteristics, laboratory characteristics, and clinical comorbidities, whereas adjustment was performed only for age and sex in 3 studies and not adjusted in 1 study; however, in these latter 4 studies, the prevalence of previous cardiovascular events and diabetes was low ($<10\%$). The quality score (range: 6–8 points) was high (Supplementary Table S1).

Demographic and main clinical characteristics of the populations examined in the included studies ($n = 39$) did not differ from those excluded because of lack of predictors ($n = 38$) (data not shown).

Obesity and development of CKD *de novo*

Overall, during a mean follow-up of 8.5 years, low eGFR occurred in 9.4% of participants (31,164 of 330,139). In

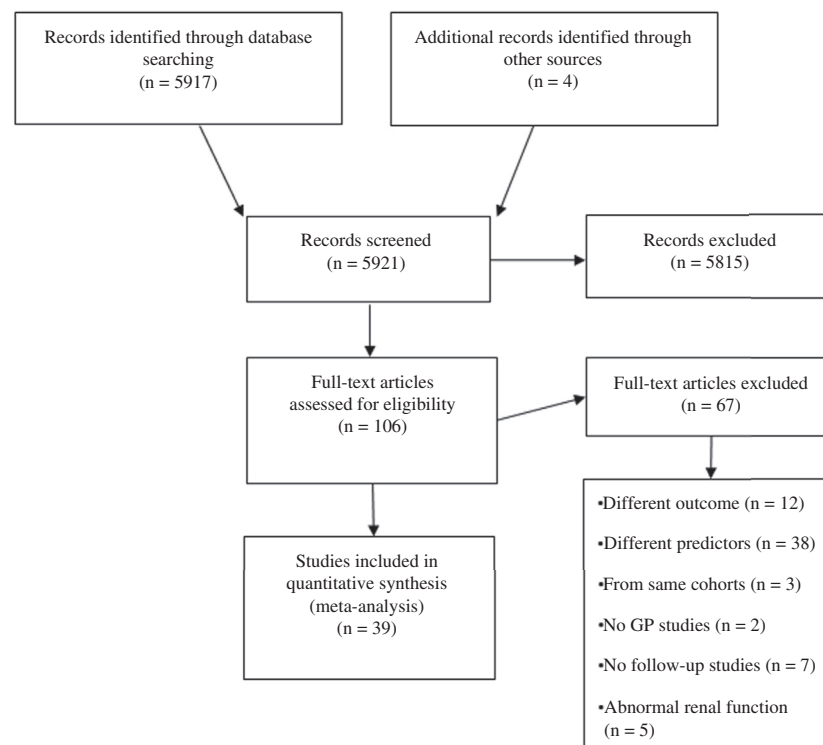


Figure 1 | Flow of selection for studies through review. GP, general population.

Table 1 | Characteristics of selected studies

Author/country	Study design	No. of subjects (% males)	Follow-up yr	Mean age	Mean BMI	Mean eGFR (formula)	Baseline albuminuria (%)	Predictors	Outcomes	Type of risk (adjustments)	Quality score
Jung ¹² /Korea	Prospective cohort	41,194 (60.5)	3.2	48.0	23.7	96.6 ^a (CKD-EPI)	NA	OB (H-U)	eGFR <60	HR (age, sex, eGFR, CVD, drinking, smoking, exercise, ALT, GGT, UA, PCR, LDL)	8
Nishikawa ¹³ /Japan	Prospective cohort	23,894 (85.9)	7.8	46.8	22.9	NA (MDRD-J)	0 (dipstick)	OB (U)	eGFR <60 UA1b	RR (age, sex, smoking, drinking, exercise, occupational exposure)	8
Watanabe ¹⁴ /Japan	Prospective cohort	34,896 (44)	5.8	59.0	22.9	104.7 (MDRD-J)	0 (dipstick)	OB	eGFR <60 UA1b	HR (age, sex)	7
Obermayr ¹⁵ /Austria	Prospective cohort	17,375 (53.6)	7.2	41.8	24.6	93.9 (MDRD)	8.6 (dipstick)	OB, OW, BMI	eGFR <60	OR (age, sex)	7
Weycker ¹⁶ /US	Retrospective cohort	54,836 (46.7)	4.5	59.2	30.0	88.4 (MDRD)	NA	OB	eGFR <60	HR (age, sex, smoking, eGFR)	8
Yamagata ¹⁷ /Japan	Prospective cohort	123,764 (33.1)	10.0	59.4	23.4	80.5 (MDRD)	0 (dipstick)	OB	eGFR <60 UA1b	HR	7
Domrongkitchaiporn ¹⁸ /Thailand	Retrospective cohort	2,967 (75.9)	12.0	42.5	23.1	NA (MDRD)	2.7 (dipstick)	OB, OW	eGFR <60	OR (BMI, SBP, DBP, DM, proteinuria, UA, TC, smoking)	8
Kanno ¹⁹ /Japan	Prospective cohort	2150 (36.6)	6.5	60.2	23.5	82.3 (MDRD-J)	0 (dipstick)	OB	eGFR <60 UA1b, CKD	HR (age, sex, smoking, drinking, SBP, CVD, DM, eGFR, TC, anti-HT treatment)	8
Tohidi ²⁰ /Iran	Prospective cohort	3313 (43.9)	9.9	39.7	26.5	74.0 (MDRD)	NA	OB, OW	eGFR <60	OR (FPG, 2hPG, TG, TC, SBP, DBP, BMI, marital status, education level, CVD, age, eGFR, WC, drugs, smoking)	8
Kawashima ²¹ /Japan	Retrospective cohort	1285 (100)	7.9	NA (> 40)	NA	NA (MDRD-J)	NA	OB	eGFR <60	HR (age, HDL, HT, DM, UA)	7
Ryu ²² /Korea	Prospective cohort	10,685 (100)	3.8	36.9	23.9	78.7 ^a (MDRD)	0 (dipstick)	OB	eGFR <60	HR (age, eGFR, GGT, UA, TG, HDL, HT, FPG)	7
Foster ²³ /US	Prospective cohort	2676 (52.3)	18.5	42.4	25.4	106.1 (MDRD)	0 (dipstick)	OB, OW	eGFR <60 ^b UA1b	OR (age, sex, DM, SBP, HT treated, smoking, HDL)	7
Gelber ²⁴ /US	Retrospective cohort	11,104 (100)	14.0	52.8	NA	NA (MDRD)	NA	OB, OW, BMI	eGFR <60	OR (age, smoking, drinking, exercise)	6
Jang ²⁶ /Korea	Retrospective cohort	53,876 (63.0)	2.0	34.9	23.2	79.9 (MDRD)	0 (dipstick)	OB, OW	UA1b	OR (sex, age, DM, HT, dyslipidemia, HDL, LDL, UA, eGFR, smoking, drinking)	7
Chang ²⁷ /US	Prospective cohort	2354 (46.7)	15.0	35.1	27.0	101.3 (CKD-EPI)	0 (ACR >3.4)	OB	UA1b ^c	OR (age, sex, race, history of KD, education, total energy intake, ACR, HT, DM)	8
Ryu ²⁸ /Korea	Prospective cohort	10,337 (100)	2.5	36.9	23.9	78.9 (MDRD)	0 (dipstick)	OB	CKD	OR (age, GFR, GGT, HDL, TG, HT)	6
Tozawa ²⁹ /Japan	Prospective cohort	6371 (63)	5.0	47.0	23.0	82.0 (MDRD-J)	0 (dipstick)	OB	CKD	OR (age, sex, smoking, drinking)	8
Malkina ³⁰ /US	Prospective cohort	4573 (47.3)	4.8	60.0	28.0	81.5 (CKD-EPI)	7 (ACR >3.4)	OW	eGFR <60	RR (age, sex, eGFR, HT, race)	8
Oh ³¹ /Korea	Prospective cohort	454 (43.0)	6.0	67.4	25.0	84.5 (MDRD)	NA	OW	eGFR <60	OR (age, sex, eGFR, smoking, education, energy intake, HT, DM, WC)	7
Ryu ³² /Korea	Prospective cohort	8792 (100)	4.0	36.9	23.8	78.7 ^a (MDRD)	NA	OW	eGFR <60 ^d	HR (age, eGFR, HDL, FPG, UA, exercise, weight change, HT)	7
Kuriyama ³³ /Japan	Retrospective cohort	8223 (77.7)	4.0	39.0	22.0	91.2 (MDRD-J)	0 (dipstick)	BMI	eGFR <60	HR (eGFR, sex, SBP, DBP, age, TC, TG, UA, HbA1c)	8
Fukui ³⁴ /Japan	Retrospective cohort	1001 (45.2)	5.7	44.6	21.5	75.2 (MDRD-J)	NA	BMI	eGFR <60	HR (age, sex, drinking, smoking, BMI, SBP, FPG, TC, UA, creatinine, Hb, K)	8

Lohr ³⁵ /US	Retrospective cohort	15,221 (98.1)	5.5	76.0	27.5	NA (CKD-EPI)	NA	BMI	eGFR <60	HR (sex, race, age, BMI, antihypertensive, COPD, cancer, DM, CVD, HDL, LDL, TG, Hb, SBP, DBP)	7
Toda ³⁶ /Japan	Prospective cohort	1388 (62.8)	5.0	53.0	22.6	71.0 (MDRD-J)	0 (dipstick)	BMI	eGFR <60	HR (age, sex, BMI, SBP, DBP, HDL, LDL, TG, HbA1c, HOMA-IR, UA)	8
McMahon ³⁷ /US	Case-control study	1321 (47.6)	10.0	59.1	27.2	82.7 ^a (CKD-EPI)	17 (dipstick)	BMI	eGFR <60	OR (age, sex)	7
Ryoo ³⁸ /Korea	Prospective cohort	18,778 (100)	4.0	41.8	24.2	81.5 (CKD-EPI)	NA	BMI	eGFR <60	OR (age, eGFR, SBP, HOMA-IR, TG, BMI, drinking, smoking, exercise, HT, DM)	7
Hyun ³⁹ /Korea	Retrospective cohort	21,859 (66.4)	5.0	39.4	23.7	77.0 (MDRD)	0 (dipstick)	BMI	eGFR <60	OR (age, sex, HT, DM, dyslipidemia, CVD, smoking, eGFR, fat mass, HDL)	8
Nakanishi ⁴⁰ /Japan	Prospective cohort	1811 (61.1)	7.7	45.5	22.6	75.8 (MDRD-J)	4 (dipstick)	BMI	eGFR <60	HR (age, sex, drinking, smoking, SBP, FPG, TC, UA, WBC, proteinuria, urine pH)	8
Sonoda ⁴¹ /Japan	Prospective cohort	7078 (64.2)	4.6	52.8	22.6	NA (MDRD-J)	0 (dipstick)	BMI	eGFR <60	OR (SBP, FPG, HDL, LDL, Hb, smoking, GFR, UA)	8
Chien ⁴² /China	Prospective cohort	5168 (63.3)	2.2	51.2	23.9	82.2 (MDRD)	4 (dipstick)	BMI	eGFR <60	HR (age, DM, DBP, stroke)	7
Tokashiki ⁴³ /Japan	Prospective cohort	28,784 (42.2)	10.0	54.0	24.1	78.5 (MDRD)	<1 (dipstick)	BMI	eGFR <60	OR (age, sex, delta BMI, SBP, DM, proteinuria)	8
Lin ⁴⁴ /China	Prospective cohort	1432 (49.3)	2.8	58.1	NA	NA (MDRD)	0 (ACR >3.4)	BMI	UA/b	OR (age, sex, smoking, drinking, exercise, SBP, DBP, CRP, FPG, eGFR)	6
Maeda ⁴⁵ /Japan	Prospective cohort	10,118 (100)	6.0	48.0	23.4	84.1 (MDRD-J)	0 (dipstick)	BMI	UA/b	HR (age, SBP, DBP, antihypertensive, DM, drinking, smoking, exercise, eGFR)	7
Komura ⁴⁶ /Japan	Prospective cohort	1506 (31.4)	10.0	58.1	22.6	NA (MDRD-J)	0 (dipstick)	BMI	CKD	OR (age, SBP, DBP, FPG, AST, GGT, serum creatinine)	8
Miyakoshi ⁴⁷ /Japan	Retrospective cohort	17,153 (59.5)	6.2	49	22.6	78.2 (MDRD-J)	0 (dipstick)	BMI	CKD	HR (sex, age, eGFR, HDL, RBC, PLT)	8
Chang ⁴⁸ /Korea	Prospective cohort	62,249 (50.5)	5	36.1	NA	80.9 (CKD-EPI)	0 (dipstick)	OB (H)	CKD	RR (sex, age, smoking, drinking, exercise, FPG, SBP, TG, HDL, LDL, CRP, ALT, AST, GGT, eGFR)	8
Mottaghi ⁴⁹ /Iran	Prospective cohort	5672 (45.2)	8.1	40.4	26.9	73.8 (MDRD)	NA	OB (H-U)	eGFR <60	HR (age, sex, smoking, HT, WC)	8
Hashimoto ⁵⁰ /Japan	Prospective cohort	3136 (41.5)	8.0	46.6	22.3	78.9 (MDRD-J)	0 (dipstick)	OB (H-U)	CKD	OR (age, sex, smoking, drinking, creatinine, UA, SBP, HDL, FPG)	8
Song ⁵¹ /Korea	Prospective cohort	1881 (37.4)	3.7	NA	NA	NA (MDRD)	NA	OB (H-U)	eGFR <60	OR (sex, drinking, smoking, exercise)	8

2hPG, 2-hours plasma glucose; ACR, albumin-creatinine ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; COPD, chronic obstructive pulmonary disease; CRP, C-reactive protein; CVD, cardiovascular disease; DBP, diastolic blood pressure; DM, diabetes mellitus; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; GGT, gamma glutamyltransferase; H, healthy obese; HR, hazard ratio; Hb, hemoglobin; HT, hypertension; KD, kidney disease; MDRD-J refers to MDRD modified formula for Japanese subjects; OB, obesity; OR, odds ratio; OW, overweight; PLT, platelets; RBC, red blood cell; RR, relative risk; SBP, systolic blood pressure; TC, total cholesterol; TG, triglycerides; TP, total protein; U, unhealthy obese evaluated as predictor; UA, uric acid; UA/b, albuminuria; WBC, white blood cells; WC, waist circumference.

In study from Chang et al.,⁴⁸ data on RR were estimated from adjusted risk difference in cumulative incidence.

^aData expressed by median.

^bIn this study outcome, was eGFR < 64 ml/min in males and < 59 ml/min in females.

^cOutcome was an ACR >2.8 mg/mmol.

^dIn this study, outcome was eGFR < 64 ml/min.

3 studies,^{12,16,21} occurrence of new-onset low eGFR during follow-up (5.2 years on average) was reported separately in obese (12.3%) and nonobese subjects (4.2%). Obesity was associated with onset of low eGFR (relative risk [RR]: 1.28; 95% confidence interval [CI]: 1.07–1.54) (Figure 2). In this pooled analysis, heterogeneity was high ($I^2 = 95.0\%$; 95% CI: 92.9–96.4; $P < 0.001$). The study by Weycker *et al.* contributed to most of the heterogeneity¹⁶; when this study was omitted, the RR was 1.18 (95% CI: 1.09–1.28), and heterogeneity decreased to 57.9% ($P = 0.006$).

Subgroup analyses are listed in Table 2. The effect was not modified after stratification for sample size, age, length of follow-up, sex, study design, reference type, baseline albuminuria, ethnicity, adjustment for eGFR, repeated measurements of eGFR, and adjustment for glycemic status and blood pressure. Subgroup categories that included the study by Weycker *et al.*¹⁶ showed a higher level of heterogeneity, whereas lower values of heterogeneity were found in the

subgroup categories that did not include the study by Weycker *et al.* For continuous variables (follow-up and age), we performed meta-regression analyses; results were not significant (Supplementary Figure S1 for age). Publication bias was not significant (Supplementary Figure S2).

Albuminuria occurred in 9.8% of participants (23,840 of 243,610) during a mean follow-up of 9.4 years. Obesity was associated with new-onset of albuminuria (RR: 1.51; 95% CI: 1.36–1.67) (Figure 2). In this pooled analysis, heterogeneity was moderate ($I^2 = 62.7\%$; 95% CI: 15.3–83.6; $P = 0.013$), with 1 study explaining most of the heterogeneity¹⁹; when this study was omitted, RR was 1.55 (95% CI: 1.42–1.70 and $I^2 = 52.2\%$; $P = 0.063$). Subgroup analyses showed that the effect was not modified after stratification for sample size, age, length of follow-up, sex, study design, reference type, ethnicity, repeated measurements of albuminuria, adjustment for glycemic status, and blood pressure (data not shown). Publication bias was not significant (Supplementary Figure S3).

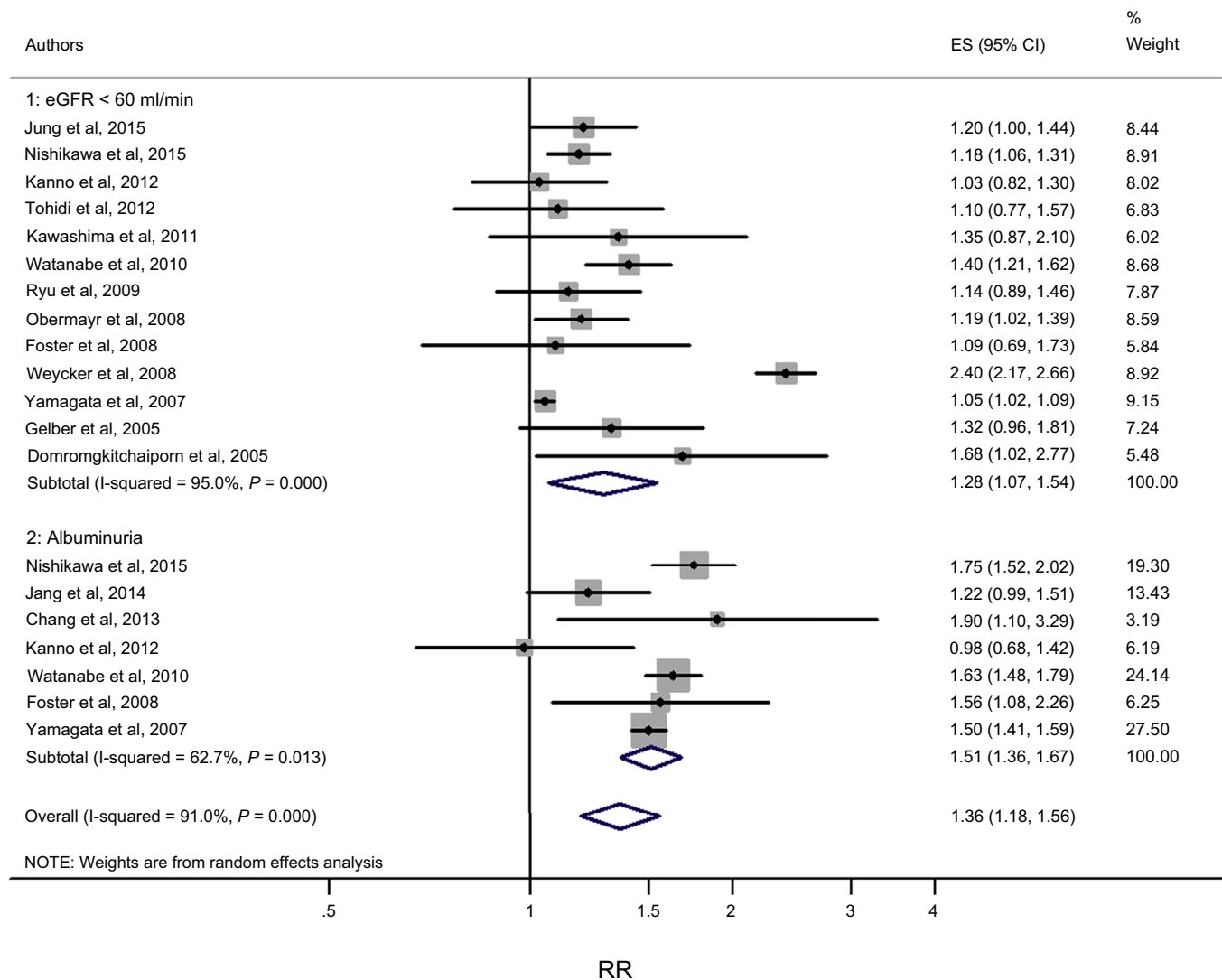


Figure 2 | Forest plot of comparison between obesity and nonobesity. Onset of decreased estimated glomerular filtration rate (eGFR) (top) and onset of albuminuria (bottom). CI, confidence interval; ES, effect size; RR, relative risk.

Table 2 | Subgroup analyses of low glomerular filtration rate onset due to obesity

Subgroup	No. of participants (cohorts)	RR (95% CI)	I ²	P
Sample size				0.317
<10,000	12,391 (5)	1.14 (0.97–1.33)	0.00	
≥10,000	317,748 (8)	1.32 (1.04–1.67)	97.03	
Age (yr)				0.477
<50	102,104 (7)	1.18 (1.10–1.27)	0.00	
≥50	226,750 (5)	1.37 (0.92–2.05)	98.03	
Follow-up (yr)				0.261
<8	186,315 (8)	1.32 (1.03–1.69)	94.62	
≥8	143,824 (5)	1.13 (0.99–1.28)	26.41	
Male sex				0.622
<50%	218,959 (5)	1.33 (0.89–1.99)	98.30	
≥50%	111,180 (8)	1.20 (1.12–1.29)	0.00	
Study design				0.786
Hazard ratio	268,810 (7)	1.32 (0.97–1.79)	97.45	
Odds ratio	37,435 (5)	1.22 (1.08–1.37)	0.00	
Reference type				0.639
Nonobese	292,704 (8)	1.30 (1.01–1.66)	97.03	
Normal weight	37,435 (5)	1.22 (1.08–1.37)	0.00	
Baseline UAlb				0.279
Present or NA	236,583 (6)	1.16 (1.05–1.29)	74.03	
Absent	93,556 (7)	1.41 (1.01–1.97)	91.88	
Ethnicity				0.489
Asian Pacific	240,835 (8)	1.18 (1.07–1.31)	69.91	
Western	89,304 (5)	1.38 (0.91–2.08)	94.46	
GFR adjustment				0.730
Yes	112,178 (5)	1.31 (0.86–2.00)	95.64	
No	217,961 (8)	1.22 (1.09–1.36)	71.51	
Repeated GFR				0.074
Yes	89,732 (2)	1.84 (1.08–3.12)	97.15	
No	240,407 (11)	1.13 (1.06–1.21)	28.73	
Glyc adjustment				0.325
Yes	20,400 (5)	1.15 (1.00–1.32)	0.00	
No	309,739 (8)	1.32 (1.03–1.68)	97.03	
BP adjustment				0.325
Yes	20,400 (5)	1.15 (1.00–1.32)	0.00	
No	309,739 (8)	1.32 (1.03–1.68)	97.03	

BP, blood pressure; CI, confidence interval; GFR, glomerular filtration rate; glyc, glycemic status; RR, relative risk; UAlb, albuminuria.

I² refers to heterogeneity among studies included in each subgroup.

P < 0.05 was significant.

Only 3 studies evaluated CKD onset as an outcome; these data are not shown due to the small number of included studies.

Summary RR, obtained by combining the renal outcomes, was 1.36 (95% CI: 1.18–1.56) (Figure 2). Sensitivity analyses that considered different correlation values (0.75 and 0.99 instead of 0.5) showed similar results (data not shown). No heterogeneity among outcomes was found ($P = 0.241$).

As additional analyses, we explored the onset of CKD in 5 studies^{12,48–51} that evaluated the risk in healthy obese subjects and 5 studies in unhealthy obese subjects.^{12,13,49–51} Both healthy and unhealthy obesity was significantly associated with an increased risk of CKD (RR: 1.30; 95% CI: 1.12–1.52 and RR: 1.63; 95% CI: 1.32–2.03, respectively) (Supplementary Figure S4), with no significant difference ($P = 0.183$).

Overweight and development of CKD *de novo*

Overweight was not a significant predictor of low eGFR (RR: 1.06; 95% CI: 0.94–1.21) (Figure 3). In this pooled analysis,

heterogeneity among studies was low ($I^2 = 50.0\%$; 95% CI: 0–77.7; $P = 0.051$). Publication bias was not significant (Supplementary Figure S5). Association of overweight and albuminuria onset was reported in only 2 studies (Figure 3). In studies that reported a risk of low eGFR related to overweight, mean follow-up duration (9.5 years) was not different from that of obesity studies. No study was identified on the relationship between overweight and CKD onset.

Summary RR, obtained by combining the 2 renal outcomes, was not significant (1.09; 95% CI: 0.98–1.21) (Figure 3). Sensitivity analyses that considered different correlation values showed similar results (data not shown). No heterogeneity among outcomes was found ($P = 0.282$).

BMI and development of CKD *de novo*

The results of pooled analysis revealed a significant positive association between BMI, as a continuous variable, and incidence of low eGFR (RR for 1 kg/m² of increment: 1.02; 95% CI: 1.01–1.03) (Figure 4). There was no significant heterogeneity among studies ($I^2 = 24.3\%$; 95% CI: 0–60.6; $P = 0.198$). Publication bias was also not significant (Supplementary Figure S6). In the only 2 studies that evaluated albuminuria onset as outcome, the association with BMI was similar to that of impaired eGFR. Only 2 cohort studies evaluated CKD onset; therefore, we could not perform a quantitative analysis.^{46,47} We performed univariate meta-regression to evaluate the association between risk of low eGFR due to an increment of 1 kg/m² BMI and mean baseline BMI. To include subjects who were more homogeneous in the analysis, we analyzed only studies on an Asian Pacific population because there were an adequate number of studies, and we found a significant increase in the effect size with an increasing mean baseline BMI (Supplementary Figure S7).

Summary RR, obtained by combining the renal outcomes, was 1.02 (95% CI: 1.01–1.03) (Figure 4). No heterogeneity among outcomes was found ($P = 0.995$).

DISCUSSION

This is the first systematic review and meta-analysis specifically aimed at evaluating the prognostic role of high BMI as predictor of new-onset CKD. Analysis included a large sample of subjects from the general adult population (>600,000) with prolonged follow-up.

In obese subjects, risk of new-onset low eGFR and albuminuria increased by 28% and 51%, respectively. This result fits well with the observed increased risk of low eGFR and albuminuria when BMI was examined as continuous variable. Conversely, overweight did not predict onset of low eGFR and albuminuria. The discrepancy with the observed significant association with incremental BMI was not readily apparent. We could hypothesize a nonlinear relationship between BMI and risk of overt CKD, that is, the risk significantly increased only in the presence of increments to higher BMI values (obesity range). In the univariate meta-regression that evaluated the risk of low eGFR associated with BMI as a

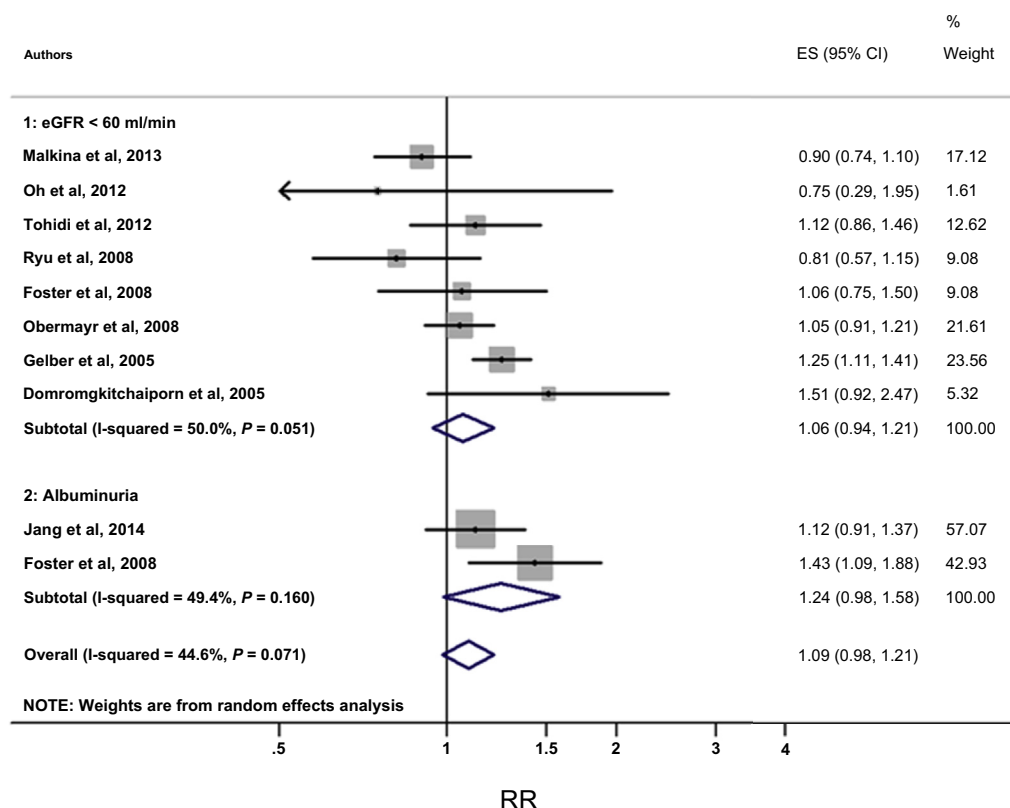


Figure 3 | Forest plot of comparison between overweight and normal weight. Outcomes: onset of decreased estimated glomerular filtration rate (eGFR) (top) and onset of albuminuria (bottom). CI, confidence interval; ES, effect size; RR, relative risk.

continuous variable (Supplementary Figure S7), we found that the effect size increased with increasing mean baseline BMI, thus supporting the hypothesis of a nonlinear relationship between BMI and risk of low eGFR. In contrast, the CKD risk associated with obesity was not modified when either normal weight subgroups or nonobese subgroups (overweight and normal weight) were used as a reference (Table 2), which suggested that the renal risk was similar in normal weight and nonobese subgroups. A similar discrepancy between the predictive role of obesity and overweight was previously reported for mortality risk.⁵² It was possible to hypothesize that compared with overweight patients, obese individuals might have less chances of receiving the best medical treatment, and at the same time, moderately increased body fat in overweight might actually be a proxy of beneficial metabolic reserves. Finally, we could not exclude that exposure to overweight for a period longer than the time span of the studies examined might increase the risk of CKD.⁵³

Our results appeared to be particularly relevant when considering that the data were extracted from large cohorts of the general population who were free of renal disease, and that estimates of risk excess were adjusted for main determinants of new-onset CKD. Furthermore, our secondary analyses showed that the risk of low eGFR related to obesity was persistent in subjects with and without metabolic syndrome.

We chose BMI to define overweight and obesity because it is the most common parameter used in clinical practice to

assess adiposity excess. In contrast, a previous meta-analysis showed that visceral fat, as assessed by waist circumference, predicted new-onset CKD. However, this meta-analysis differed from the present analysis, because BMI was evaluated in only 3 studies.⁵⁴

The mechanisms by which obesity may initiate and exacerbate CKD remain elusive. Abnormality in renal hemodynamics is probably the main factor; however, a contribution of inflammatory and metabolic mechanisms can be hypothesized.⁵⁵

In particular, studies in animals and in humans have shown that obesity is associated with elevated GFR and increased renal blood flow.^{56,57} This likely occurs because afferent arteriolar dilation due to proximal salt reabsorption, coupled with efferent renal arteriolar vasoconstriction, is dependent on elevated angiotensin II generation.^{55,58} These effects may contribute to hyperfiltration, glomerular hypertrophy, focal glomerulosclerosis, and, in turn, to proteinuria. Furthermore, interference with renal blood flow as a result of compression of the renal hilum and/or renal parenchyma by visceral fat, other hemodynamic factors (high intra-abdominal pressure or cardiopulmonary dysfunction), and synthesis of several proteins of the renin-angiotensin system (RAS) by visceral fat may play a role.^{55,58–60} Finally, abnormal activation of RAS, which occurs despite sodium retention and expansion of extracellular fluid volume,^{59,60} is common in obese individuals and may directly account for glomerular hypertension and onset of proteinuria.

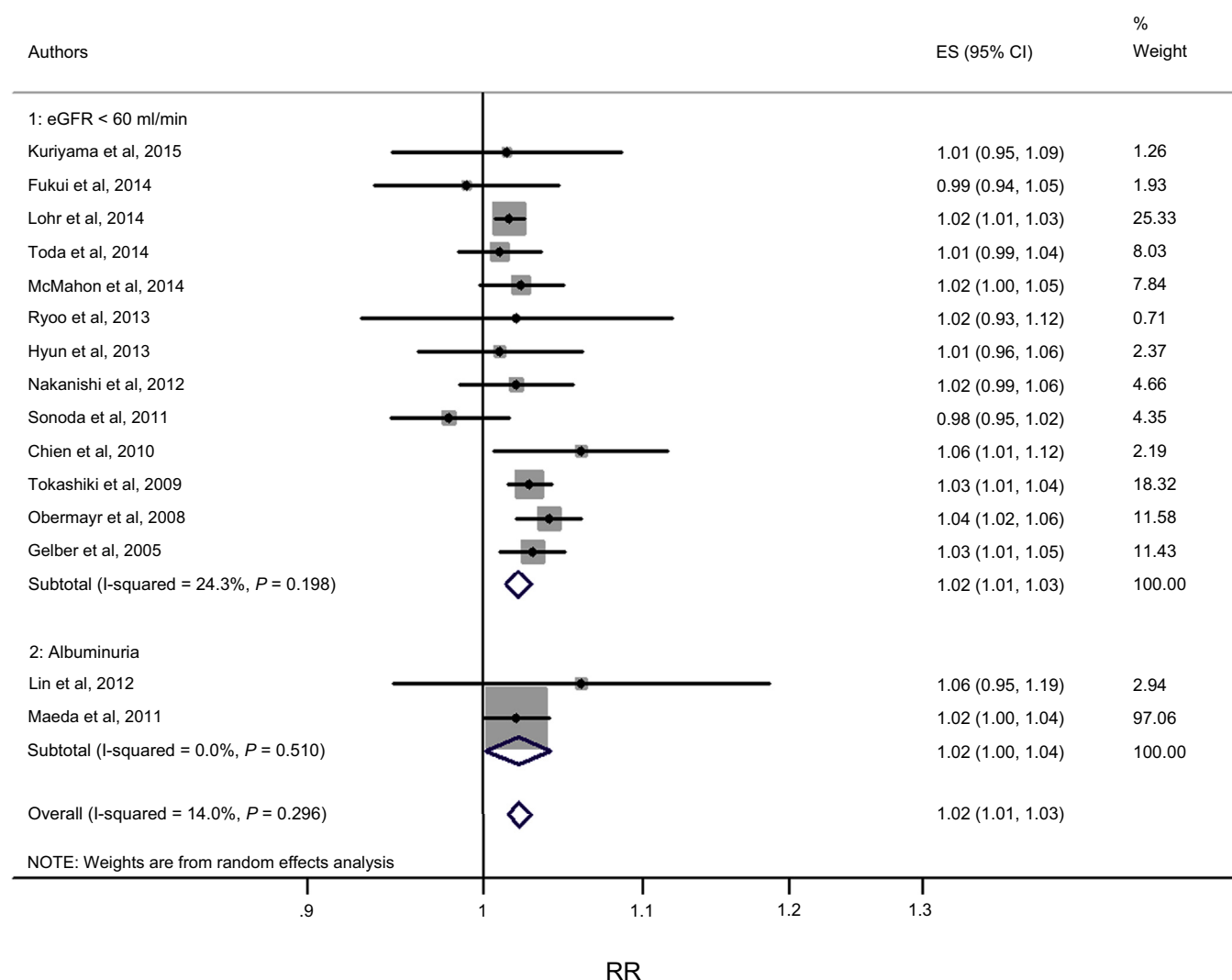


Figure 4 | Forest plot of comparison for every increment of 1 kg/m² of body mass index. Outcomes: onset of decreased estimated glomerular filtration rate (eGFR) (top) and onset of albuminuria (bottom). CI, confidence interval; ES, effect size; RR, relative risk.

Interestingly, the role of RAS might persist in overt CKD; in agreement with this hypothesis, a *post hoc* analysis of the Ramipril Efficacy In Nephropathy (REIN) study in CKD patients showed that use of RAS inhibitors allowed a greater nephroprotection in obese versus nonobese individuals.⁶¹

We found that obesity was a significant predictor of CKD *de novo* not only in terms of eGFR decline but also when considering new-onset albuminuria. Due to the large prevalence of CKD and the higher cardiovascular risk heralded by this condition *per se*,^{4–6} weight loss intervention in obese subjects could be therefore considered as a main strategy to limit the CKD burden in the general population. The importance of this preventive approach grows when considering that once CKD has developed, the role of abnormal BMI on progression of CKD becomes less evident.^{62–64}

Our meta-analysis had some limitations. First, no access to individual patient-level data was available. However, data extracted from studies and pooled in meta-analysis were adjusted for co-morbidities. Furthermore, the high quality

score of included studies gave strength to the results. Second, the occurrence of outcomes might be overestimated because it was based on a single assessment in most studies; however, the lack of difference in the comparison between studies with and without repeated measurements of outcome (chronicity criterion) allowed us to reasonably exclude the possibility of acute renal impairment. Third, most studies did not report mortality data; therefore, we could not exclude the potential effect of survival bias and competing risk. Fourth, although analyses were adjusted for multiple determinants of CKD *de novo*, the effect of residual confounders could not be excluded. In particular, the high heterogeneity found among studies that evaluated the risk of low GFR related to obesity could be explained by genetic and environmental factors that were not assessed in the included studies. Furthermore, the different heterogeneity in the subgroup analyses was mainly dependent on 1 study,¹⁶ likely because of the high prevalence of cardiovascular comorbidities in that study compared with the other studies included in our meta-analysis. Finally,

although BMI could affect the level of renal function calculated by Cockcroft-Gault formula, bias became irrelevant when GFR was estimated by the equations used in the examined studies included in this meta-analysis.⁶⁵

In conclusion, this systematic review and meta-analysis suggests that obesity acts as independent predictor of new-onset CKD, and this association persists for either albuminuric stages (1–2) without renal insufficiency and stages 3 or more (GFR <60) of CKD. Renal risk is limited to the status of obesity while not being manifest for the overweight condition and also persists independently from the presence of metabolic syndrome. These findings may have important clinical implications in terms of risk stratification and preventive strategies, especially when considering that obesity and CKD are now considered 2 of the main epidemic diseases worldwide.

METHODS

Search strategies

This review was carried out in accordance with the Meta-analysis of Observational Studies in Epidemiology guidelines (Supplementary Item 1).^{66,67} A systematic search of articles published without language restriction was performed using PubMed and ISI Web of Science databases to identify relevant studies published from January 1, 2000 to August 31, 2016. Date restriction was essential to standardize studies in terms of the definition of CKD based on Kidney Disease Outcomes Quality Initiative guidelines published in the early 2000s, with eGFR and albuminuria used as markers of severity of kidney disease. The following Medical Subject Headings and text words were used: “obesity,” “body mass index,” “overweight,” “BMI,” “over-weight,” “abdominal obesity,” “chronic renal insufficiency,” “proteinuria,” “albuminuria,” “chronic kidney disease,” “chronic kidney failure,” “chronic kidney insufficiency,” “chronic kidney dysfunction,” “chronic renal failure,” “chronic renal dysfunction,” “kidney disease,” “CKD,” “risk factors,” “risk factor,” “incidence,” “incident,” “development,” “new-onset,” “predictors,” and “predictor.” Further details are reported in Supplementary Item 2. Bibliographies of relevant articles and reviews were manually screened for identify additional studies.

Study selection and quality assessment

Criteria for inclusion were: (i) observational longitudinal studies of adult subjects (age older than 18 years) from the general population; (ii) normal kidney function at baseline, defined as eGFR ≥ 60 ml/min/1.73 m², and, where indicated, negative urinary dipstick or albumin/creatinine ratio <3.4 mg/mmol; and (iii) reported risk of new-onset CKD, isolated decreased eGFR, or albuminuria, expressed as either hazard ratio (HR), odds ratio (OR), or RR with 95% (CI) related to obesity or overweight or as a continuous BMI value. We also selected studies that evaluated the risk of new-onset CKD in healthy versus unhealthy obesity as an additional analysis. We collected data from the most adjusted model in each study. In the case of overlapping studies in the same cohort, we only examined those with longer follow-up. We excluded studies published as abstracts, letters to editors, commentaries, and reviews.

Two investigators (CG and SB) independently obtained the full reports of potentially relevant studies and reviewed each study using

predefined eligibility criteria. Any discrepancy between the 2 reviewers on study eligibility was resolved by consensus agreement.

The Newcastle-Ottawa Scale was used for quality assessment (Supplementary Table S1).⁶⁸ Risk of bias was assessed among the cohort studies by considering 3 items: (i) selection of participants (containing 4 domains); (ii) comparability (1 domain); and (iii) outcome measure (containing 3 domains). Each domain was rated as “yes,” “no,” or “unclear.” Each quality domain was categorized as low risk for bias (yes) when the study reported adequate data and met the criteria, and as high risk (no) when the study reported adequate data but did not meet the criteria for the specific quality domain. Studies that did not report data to allow quality assessment were categorized as “unclear” and, therefore, potentially carried a high risk of bias. “Yes” was scored as 1 and “no” or “unclear” were scored as zero. A quality bar was plotted for each domain to examine the limitations of the studies. High-quality studies were defined by a score of >5 points. Disagreements in the scores were resolved by discussion and consensus between the 2 reviewers.

Predictors and outcomes

Predictors were obesity, overweight, and BMI level as continuous variables. According to World Health Organization, obesity, overweight and normal BMI were defined in the Western population as BMI values of >30, 25 to 30, and <25 kg/m², respectively; for the Asian Pacific population, the recommended cutoff values of BMI corresponding to these 3 categories were >25, 23 to 25, and <23 kg/m², respectively. An additional predictor of interest was continuous BMI, with the renal risk estimated for an increment of 1 kg/m². Unhealthy obesity referred to obese patients with metabolic syndrome, defined according to National Cholesterol Education Program-Adult Treatment Panel III criteria.

According Kidney Disease Outcomes Quality Initiative guidelines,⁶⁹ the 2 main indicators of renal function are eGFR and albuminuria. Therefore, we identified 3 outcomes: (i) isolated low eGFR (<60 ml/min/1.73 m²); (ii) isolated albuminuria (dipstick $\geq 1+$ or albumin/creatinine ratio ≥ 3.4 mg/mmol); and (iii) CKD, defined by the presence of either low eGFR or albuminuria.

Statistical analysis

We quantified the inter-rater agreement for study selection and quality assessment. Extracted results on risk estimate (HR, OR, RR) were pooled in the meta-analysis separately for CKD, isolated decreased eGFR, and albuminuria onset. Increments of BMI different from 1 unit (kg/m²) were standardized to the same pre-established increment. We assumed a conservative approach in pooling results by using a random-effects model, which allowed for variation of true effects across studies. The summary estimate was given as RR, which could be interpreted approximately as OR and HR. Heterogeneity was analyzed using a χ^2 test on $N - 1$ degrees of freedom and with the I^2 test with 95% CIs.⁷⁰ I^2 values of 25%, 50%, and 75% corresponded to cutoff points for low, moderate, and high degrees of heterogeneity.

To exclude the possibility that a study exerted excessive influence on the heterogeneity, we conducted a sensitivity analysis by omitting 1 study at a time and then reanalyzing the data to assess the change in effect size.⁷¹ Moreover, to explore possible sources of heterogeneity (where significant), we performed univariate random-effects meta-regression and moderator analyses. Meta-regression was used to test differences between moderators. Restricted maximum likelihood estimators were used to estimate model parameters.⁷² Analysis of moderators included sample size ($\geq 10,000$ vs. <10 000), age

(50 years of age or older vs. younger than 50 years of age), length of follow-up (≥ 8 vs. < 8 years), male sex (male prevalence $\geq 50\%$ of cohort vs. $< 50\%$ of cohort), ethnicity (Asian Pacific vs. Western population), reference type (overweight combined with normal weight vs. normal weight), presence of albuminuria at baseline (present or unavailable vs. no albuminuria), repeated measurements of eGFR and albuminuria (at least 3-month interval) to confirm outcome persistence during follow-up (present vs. absent), adjustment for baseline eGFR, blood pressure, and glycemic status (present vs. absent). Additional subgroup analysis was performed to investigate differences in the measures used for risk estimation (OR and HR). When OR was used as main measure with an overall risk of outcome $> 10\%$, to verify the risk of overestimating an association, we performed a sensitivity analysis that excluded these studies. Publication bias was assessed by funnel plot, and the Begg and Egger test.⁷³ Summary estimates combining the 3 previously mentioned renal outcomes were calculated; when multiple outcomes were analyzed in a particular study, the method of Borenstein *et al.* was used, which considered a correlation between estimates of 0.5.⁷⁴ Sensitivity analyses that considered a correlation of 0.75 and 0.99 were also performed. Subgroup analysis and meta-regression were used to test heterogeneity of estimates among outcomes. For subanalyses on healthy and unhealthy obesity, the 2 endpoints used in the studies (low eGFR in 4 and CKD in 2) were considered as a single comprehensive endpoint (CKD). Statistical significance was defined as a 95% CI with no overlap with the null effect value. A 2-sided *P* value < 0.05 was considered significant.

Analyses were performed using PROMETA 2 (INTERNOVI, Cesena, Italy) and Stata/SE 11 (Stata Corporation, College Station, Texas).

DISCLOSURE

All the authors declared no competing interests. This work was endorsed by the Italian Society of Nephrology (Gruppo di Studio sul Trattamento Conservativo della Insufficienza Renale Cronica) without any financial support. There was no funding source for this study.

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Author contributions

Study concept and design: CG, RM, LDN, and GC. Acquisition, analysis, or interpretation of data: CG, SB, PC, RM, LDN, and GC. Drafting of the manuscript: CG, SB, and GC. Critical revision of the manuscript for important intellectual content: RM, LDN, and GC. Statistical analysis: CG, PC, and SB. Study supervision: RM, LDN, and GC. Each author contributed important intellectual content during drafting of the manuscript or revisions, and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved. CG and GC have full access to data and take responsibility that this study has been reported honestly, accurately, and transparently; that no important aspects of the study have been omitted and that any discrepancies from the study as planned have been explained.

SUPPLEMENTARY MATERIALS

Item S1. The Meta-analysis of Observational Studies in Epidemiology (MOOSE) checklist of the article.

Item S2. Literature search strategy for PubMed.

Table S1. Newcastle-Ottawa Scale (NOS) score for evaluation of study quality.

Figure S1. Meta-regression on the relationship between increasing age and risk for decreased estimated glomerular filtration rate among studies that evaluated obesity as a predictor.

Figure S2. Funnel plot for publication bias among studies that evaluated association obesity and decreased glomerular filtration rate (GFR).

Figure S3. Funnel plot for publication bias among studies that evaluated the association between obesity and albuminuria onset.

Figure S4. Forest plot of comparison between healthy obese (top) and unhealthy obese versus normal weight for the following outcome: onset of kidney disease. CI, confidence interval; ES, effect size; RR, relative risk; W, weight.

Figure S5. Funnel plot for publication bias among studies evaluating association overweight and decreased glomerular filtration rate (GFR).

Figure S6. Funnel plot for publication bias among studies evaluating association of 1 kg/m² increment of body mass index (BMI) and decreased glomerular filtration rate (GFR).

Figure S7. Meta-regression on the relationship between increasing baseline body mass index (BMI) and risk for decreased estimated glomerular filtration rate among studies evaluating 1 kg/m² as a predictor. Effect size represents measure of risk associated with increasing 1 kg/m² of BMI. *P* = 0.045.

Supplementary material is linked to the online version of the paper at www.kidney-international.org.

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